

Experiment no:05

Name:Dayanand Jagdale Roll no:42

Init.py all=

```
['index','join','login','logout','list','write']
```

Index.py

```
from flask import render_template from
board.board_blueprint import board
@board.route('/') def
index():
return render_template(index.html, title=Index Page)
```

join.py

```
import pymysql from flask import render_template,
request, current_app from board.board_blueprint
import board
@board.route('/join') def
join():
return render_template(join.html)
@board.route('/join_process,
methods=[POST]) def join_process():
id = request.form[id] password =
request.form[password_1] email =
```

```
request.form[email] print(id:[%s]
password:[%s] email:[%s] % (id,
password, email)) db_address =
current_app.config[DB_ADDRESS]
db_port =
current_app.config[DB_PORT] db_id =
current_app.config[DB_ID]
db_password =
current_app.config[DB_PASSWORD]
db_name =
current_app.config[DB_NAME] conn =
pymysql.connect(host=db_address,
port=int(db_port), user=db_id,
password=db_password,
db=db_name,
charset=utf8)
try:
    cursor = conn.cursor() sql = quotINSERT INTO users(id, password, email) VALUES(%s, %s,
%s)quot % (id, password, email) cursor.execute(sql) conn.commit()
finally:
    conn.close() return render_template(&join.html&,
title=&Member Join&)
```

list.py

```
import pymysql from flask import render_template,
request, current_app from board.board_blueprint
import board

@board.route(/list, methods=[GET])
def list():
    page = request.args.get(page) db_address =
    current_app.config[DB_ADDRESS] db_port =
    current_app.config[DB_PORT] db_id =
    current_app.config[DB_ID] db_password =
    current_app.config[DB_PASSWORD] db_name =
    current_app.config[DB_NAME] conn =
    pymysql.connect(host=db_address,
    port=int(db_port), user=db_id,
    password=db_password, db=db_name,
    charset='utf8') try:
    cursor = conn.cursor() sql = quotSELECT `no`, `content`, `writer`, `read` FROM board ORDER BY
    `write_time` DESCquot cursor.execute(sql) rows = cursor.fetchall() for row_data in rows:
    print(no:[%s] content:[%s] writer:[%s] read:[%s] % (row_data[0], row_data[1], row_data[2],
    row_data[3]))
    finally:
    conn.close() return render_template(list.html, rows=rows,
    title=Article List) login.py
```

```

import pymysql from flask import render_template, request, current_app, session,
redirect, url_for from functools import wraps from board.board_blueprint import
board from board.board_logger import Log def login_required(f): @wraps(f) def
decorated_function(*args, **kwargs):
try:
session_key = request.cookies.get(current_app.config[SESSION_COOKIE_NAME])
print(session_key: [%s] & % session_key) is_login = False if session.sid ==
session_key and session.contains (usr):
is_login = True if
not is_login:
return redirect(url_for(.login_form, next=request.url))
return f(*args, **kwargs) except Exception as e:
Log.error(Login error : %s % str(e))
return decorated_function
@board.route(/login) def
login_page():
next_url = request.args.get(next, )
id = request.args.get(id, ) password =
request.args.get(password, ) return
render_template(login.html)
@board.route(/login, methods=[POST])
def login_process():
next_url = request.args.get(next) id =
request.form[id] password =
request.form[password] db_address =
current_app.config[DB_ADDRESS] db_port =

```

```

current_app.config[DB_PORT] db_id =
current_app.config[DB_ID] db_password =
current_app.config[DB_PASSWORD] db_name =
current_app.config[DB_NAME] conn =
pymysql.connect(host=db_address,
port=int(db_port), user=db_id,
password=db_password, db=db_name,
charset='utf8') try:
cursor = conn.cursor() sql = quotSELECT `usn`, `id`, `email`, `update_time` FROM
users WHERE `id`=%s AND
`password`=%s LIMIT 1quot % (id, password)
print(sql) cursor.execute(sql) rows =
cursor.fetchall()
if rows: for row_data in rows: session.permanent = True usn = row_data[0] id =
row_data[1] email = row_data[2] update_time = row_data[3] print(usn:[%s] id:[%s]
email:[%s] update_time:[%s] % (usn, id, email, update_time)) session[usn] = usn
session[user] = id session[email] = email if next_url != and next_url != None:
return redirect(url_for(next_url)) else: return redirect(url_for(.list)) else:
print('Cannot found user')
finally:
conn.close() return render_template(39login.html39, next_url=next_url,
title=39Member Login39)

```

logout.py

```

from flask import session, redirect, url_for from
board.board_blueprint import board
@board.route(/logout) def logout():
session.clear() return redirect(url_for(.list))

```

write.py

```

import pymysql from flask import render_template, request, redirect,
url_for, current_app from board.board_blueprint import board
@board.route(/write, methods=[GET, POST]) def write(): if
request.method == POST: writer = request.form[writer] content =
request.form[content] if writer and content:
print(POST writer:[%s] content:[%s] % (writer, content)) else:
return render_template(write.html) db_address =
current_app.config[DB_ADDRESS] db_port =
current_app.config[DB_PORT] db_id =
current_app.config[DB_ID] db_password =
current_app.config[DB_PASSWORD] db_name =
current_app.config[DB_NAME] conn =
pymysql.connect(host=db_address,
port=int(db_port),
user=db_id, password=db_password,
db=db_name,
charset=utf8)
try:

```

```
cursor = conn.cursor() sql = quotINSERT INTO board(writer, content) VALUES(%s,
%s)quot % (writer, content) print(sql) cursor.execute(sql) conn.commit()
finally:
conn.close() return redirect(url_for(.list)) return
render_template(write.html, title=Article Write)
```

Board_blueprint.py

```
from flask import Blueprint board = Blueprint(bikeparking, name,
template_folder=../templates, static_folder=../static)
```

board_config.py class

```
FlaskBoardConfig(object):
```

```
DB_ADDRESS = "
```

```
DB_PORT = "
```

```
DB_ID = "
```

```
DB_PASSWORD ="
```

```
DB_NAME = "
```

Board_logger.py

```
import logging from logging import getLogger,
```

```
handlers, Formatter class Log:
```

```
log_level_map = { debug
```

```
: logging.DEBUG, info :
```

```
logging.INFO, warn :
```

```

logging.WARN, error :
logging.ERROR, critical :
logging.CRITICAL
}

my_logger = None @staticmethod def init(logger_name=flaskboardlog,
log_level=debug, log_filepath=board/logs/log.log):
Log. my_logger = getLogger(logger_name)
Log.my_logger.setLevel(Log.log_level_map.get(log_level, warn))
formatter = Formatter(%(asctime)s - %(levelname)s - %(message)s)
console_handler = logging.StreamHandler()
console_handler.setFormatter(formatter) Log.
my_logger.addHandler(console_handler) file_handler = \
handlers.TimedRotatingFileHandler(log_filepath, when=D, interval=1)
file_handler.setFormatter(formatter)

Log. my_logger.addHandler(file_handler)

@staticmethod def debug(msg):
Log. my_logger.debug(msg)

@staticmethod def
info(msg):
Log. my_logger.info(msg)

@staticmethod def
warn(msg):
Log. my_logger.warn(msg)

@staticmethod def
error(msg):

```



```
Log. my_logger.error(msg) @staticmethod  
def critical(msg): Log. my_logger.critical(msg)
```

Output:



Result : Thus the implementation of Bulletin Board was successfully done

Experiment no:06

Name:Dayanand Jagdale Roll No:42

```
package PackageDemo; import java.io.IOException; import
org.apache.hadoop.conf.Configuration; import
org.apache.hadoop.fs.Path; import
org.apache.hadoop.io.IntWritable; import
org.apache.hadoop.io.LongWritable; import
org.apache.hadoop.io.Text; import
org.apache.hadoop.mapreduce.Job; import
org.apache.hadoop.mapreduce.Mapper; import
org.apache.hadoop.mapreduce.Reducer; import
org.apache.hadoop.mapreduce.lib.input.FileInputFormat; import
org.apache.hadoop.mapreduce.lib.output.FileOutputFormat; import
org.apache.hadoop.util.GenericOptionsParser; public class
WordCount { public static void main(String [] args) throws Exception
{
Configuration c=new Configuration();
String[] files=new GenericOptionsParser(c,args).getRemainingArgs();
Path input=new Path(files[0]);
Path output=new Path(files[1]); Job j=new
Job(c,"wordcount");
j.setJarByClass(WordCount.class);
j.setMapperClass(MapForWordCount.class);
j.setReducerClass(ReduceForWordCount.class);
j.setOutputKeyClass(Text.class);
```

```

j.setOutputValueClass(IntWritable.class); FileInputFormat.addInputPath(j,
input);
FileOutputFormat.setOutputPath(j, output);
System.exit(j.waitForCompletion(true)?0:1);
}

```

```

public static class MapForWordCount extends Mapper<LongWritable, Text, Text,
IntWritable>{ public void map(LongWritable key, Text value, Context con)
throws IOException,
InterruptedException

```

```

{
    String line = value.toString();
    String[]
words=line.split(",");
for(String word: words )
{
    Text outputKey = new Text(word.toUpperCase().trim()); IntWritable
outputValue = new IntWritable(1); con.write(outputKey,
outputValue);
}
}
}

```

```

public static class ReduceForWordCount extends Reducer<Text, IntWritable, Text,
IntWritable>
{
    public void reduce(Text word, Iterable<IntWritable> values, Context con) throws
IOException,
InterruptedException

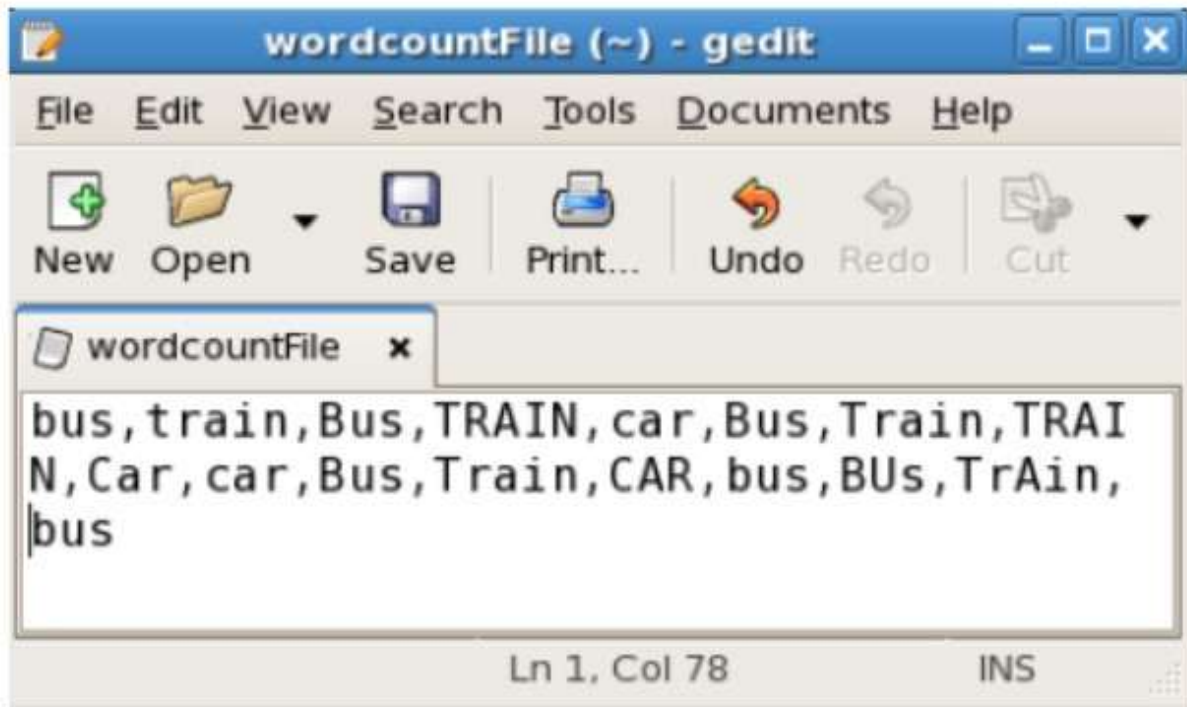
```

```

{
int sum = 0; for(IntWritable
value : values)
{
sum += value.get();
}
con.write(word, new IntWritable(sum));
}
}
}

```

Output:



To move this into Hadoop directly, open the terminal and enter the following commands:

```
[training@localhost ~]$ hadoop fs -put wordcountFile wordCountFile
```

2. Run the jar file:

(Hadoop jar jarfilename.jar packageName.ClassName PathToInputTextFile
PathToOutputDirectory)

```
[training@localhost ~]$ hadoop jar MRProgramsDemo.jar PackageDemo.WordCount  
wordCountFile MRDir1
```

3. Open the result:

```
[training@localhost ~]$ hadoop fs -ls MRDir1
```

Found 3 items

```
-rw-r--r-- 1 training supergroup 0 2042-02-23 03:36 /user/training/MRDir1/_SUCCESS drwxr-xr-  
x - training supergroup 0 2042-02-23 03:36 /user/training/MRDir1/_logs  
-rw-r--r-- 1 training supergroup 20 2042-02-23 03:36 /user/training/MRDir1/part-r-00000
```

```
[training@localhost ~]$ hadoop fs -cat MRDir1/part-r-00000
```

BUS 7

CAR 4

TRAIN 6

Result: A word count application which counts the number of occurrences of each word a large collection of documents Using Map Reduce model was successfully developed.

Experiment no:07

Name:Dayanand Jagdale Roll no:42

The following code for the Board class:

```
package com.ssaurel.tictactoe import
java.util.Random public class Board { private static
final Random RANDOM = new Random() private
char[] es private char currentPlayer private boolean
ended public Board() { es = new char[9] newGame()
}
public boolean isEnded() { return
ended
}
public char play(int x, int y) { if (!ended
"amp"amp es[3 * y + x] == "" "") { es[3 * y +
x] = currentPlayer changePlayer()
}
return checkEnd()
}
public void changePlayer() { currentPlayer =
(currentPlayer == "X" ? "O" : "X")
}
public char getElt(int x, int y) { return
elts[3 * y + x]
}
```

```

public void newGame() {
    for (int i = 0; i < elts.length; i++) { elts[i]
        = "" }
    currentPlayer = "X"
    ended
        = false
    }

    public char checkEnd() {
        for (int i = 0; i < 3; i++) { if (getElt(i,
            0) != "" & getElt(i, 0) ==
            getElt(i, 1) & getElt(i, 1) ==
            getElt(i, 2)) { ended = true; return
            getElt(i, 0)
        }
        if (getElt(0, i) != "" &
            getElt(0, i) == getElt(1, i) &
            getElt(1, i) == getElt(2, i)) { ended =
            true; return getElt(0, i)
        }
    }

    if (getElt(0, 0) != "" &
        getElt(0, 0) == getElt(1, 1) &
        getElt(1, 1) == getElt(2, 2)) { ended =
        true; return getElt(0, 0)
    }
}

```



```

if (getElt(2, 0) != "" "" "amp"amp
getElt(2, 0) == getElt(1, 1) "amp"amp
getElt(1, 1) == getElt(0, 2)) { ended =
true; return getElt(2, 0);
} for (int i = 0; i lt; 9; i++)
{ if (elts[i] == ""; "");)
return ""; "";;;
}
return "";T"";
}

public char computer() { if
(!ended) { int position = -1; do
{ position =
RANDOM.nextInt(9);
} while (elts[position] != ""; ""); elts[position] = currentPlayer; changePlayer();
}
return checkEnd();
}
}

```

Rendering the Board on the Screen

The following code for the BoardView object :

```

package com.ssaurel.tictactoe; import
android.content.Context; import

```

```
android.graphics.Canvas; import
android.graphics.Color; import
android.graphics.Paint; import
android.support.annotation.Nullable; import
android.util.AttributeSet; import
android.view.MotionEvent; import
android.view.View; public class BoardView
extends View { private static final int
LINE_THICK = 5; private static final int
ELT_MARGIN = 20; private static final int
ELT_STROKE_WIDTH = 15; private int width,
height, eltW, eltH; private Paint gridPaint,
oPaint, xPaint;

private GameEngine gameEngine;
private MainActivity activity; public
BoardView(Context context) {
super(context);
}

public BoardView(Context context, @Nullable AttributeSet attrs) {
super(context, attrs); gridPaint = new Paint(); oPaint = new
Paint(Paint.ANTI_ALIAS_FLAG); oPaint.setColor(Color.RED);
oPaint.setStyle(Paint.Style.STROKE);
oPaint.setStrokeWidth(ELT_STROKE_WIDTH); xPaint = new
Paint(oPaint); xPaint.setColor(Color.BLUE);
}
```

```

public void setMainActivity(MainActivity a) { activity
= a;
}

public void setGameEngine(GameEngine g) { gameEngine
= g;
}

@Override protected void onMeasure(int
widthMeasureSpec, int

height = View.MeasureSpec.getSize(heightMeasureSpec); width
= View.MeasureSpec.getSize(widthMeasureSpec); eltW = (width
- LINE_THICK) / 3; eltH = (height - LINE_THICK) / 3;
setMeasuredDimension(width, height);
}

@Override protected void
onDraw(Canvas canvas) {
drawGrid(canvas); drawBoard(canvas);
}

@Override public boolean onTouchEvent(MotionEvent
event) { if (!gameEngine.isEnded() &&
event.getAction() == N) { int x = (int) (event.getX() /
eltW); int y = (int) (event.getY() / eltH); char win =
gameEngine.play(x, y); invalidate(); if (win != "";) {
activity.gameEnded(win);
} else {
// computer plays ...

```

```

win = gameEngine.computer();
invalidate(); if
(win != ""; "") {
activity.gameEnded(win);
}
}
}
return super.onTouchEvent(event);
}

private void drawBoard(Canvas canvas) {
for (int i = 0 i < 3 i++) { for (int j = 0 j < 3
j++) { drawElt(canvas, gameEngine.elts(i,
j), i, j)
}
}
}

private void drawGrid(Canvas canvas) {
for (int i = 0 i < 2 i++) { // vertical lines float left =
eltsW * (i + 1) float right = left + LINE_THICK float
top = 0 float bottom = height
canvas.drawRect(left, top, right, bottom, gridPaint)
// horizontal lines float left2 = 0 float right2 = width
float top2 = eltsH * (i + 1) float bottom2 = top2 +
LINE_THICK canvas.drawRect(left2, top2, right2,
bottom2, gridPaint)
}
}

```

```

}

private void drawElt(Canvas canvas, char c, int x, int y) { if (c ==
""O""") { float cx = (eltW * x) + eltW / 2 float cy = (eltH * y) + eltH /
2 canvas.drawCircle(cx, cy, Math.min(eltW, eltH) / 2 - ELT_MARGIN
*

} else if (c == ""X""") { float startX = (eltW * x) +
ELT_MARGIN float startY = (eltH * y) + ELT_MARGIN
float endX = startX + eltW - ELT_MARGIN * 2 float endY
= startY + eltH - ELT_MARGIN canvas.drawLine(startX,
startY, endX, endY, xPaint) float startX2 = (eltW * (x + 1))
- ELT_MARGIN float startY2 = (eltH * y) + ELT_MARGIN;
float endX2 = startX2 - eltW + ELT_MARGIN * 2; float
endY2 = startY2 + eltH - ELT_MARGIN;
canvas.drawLine(startX2, startY2, endX2, endY2, xPaint);
}
}
}

```

Creating the UI for our Game

```

<?xml version="1.0" encoding="utf-8"?>

<RelativeLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent" tools:context="com.ssaurel.tictactoe.MainActivity">

```

```
<com.ssaurel.tictactoe.BoardView android:id="@+id/board"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:layout_centerInParent="true"/>
</RelativeLayout>
```

Starting a new Game

```
<?xml version="1.0" encoding="utf-8"?>
<menu xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto">
    <item android:id="@+id/action_new_game"
        android:orderInCategory="50"
        android:title="New Game"
        app:showAsAction="always" />
</menu>
```

Assemble all the pieces of the puzzle

```
package com.ssaurel.tictactoe; import
    android.content.DialogInterface; import
    android.os.Bundle; import
    android.support.v7.app.AlertDialog; import
    android.support.v7.app.AppCompatActivity; import
    android.view.Menu; import android.view.MenuItem;
import static com.ssaurel.tictactoe.R.id.board; public
```

```

class MainActivity extends AppCompatActivity {
    private BoardView boardView; private GameEngine
    gameEngine;

    @Override protected void onCreate(Bundle
    savedInstanceState) {
        super.onCreate(savedInstanceState);

        setContentView(R.layout.activity_main); boardView =
        (BoardView) findViewById(board); gameEngine = new
        GameEngine();

        boardView.setGameEngine(gameEngine);

        boardView.setMainActivity(this);
    }

    @Override public boolean
    onCreateOptionsMenu(Menu menu) {
        getMenuInflater().inflate(R.menu.main, menu);

        return super.onCreateOptionsMenu(menu);
    }

    @Override
    public boolean onOptionsItemSelected(MenuItem item) { if
    (item.getItemId() == R.id.action_new_game) { newGame();
    }

    return super.onOptionsItemSelected(item);
    }

    public void gameEnded(char c) {

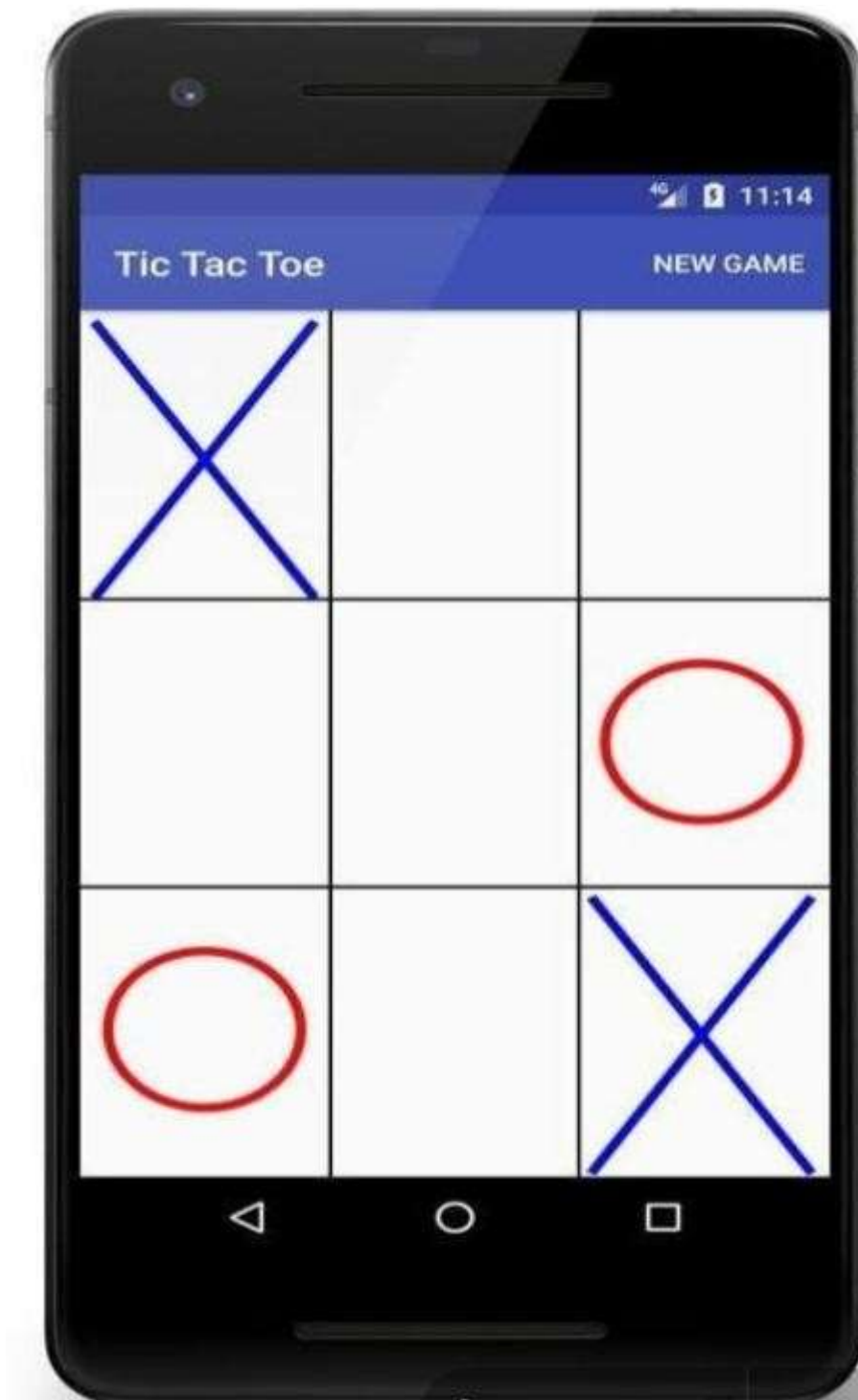
        String msg = (c == 'T') ? "Game Ended. Tie" : "GameEnded. " + c + " win";;

        new AlertDialog.Builder(this).setTitle("Tic Tac Toe");. setMessage(msg).

```

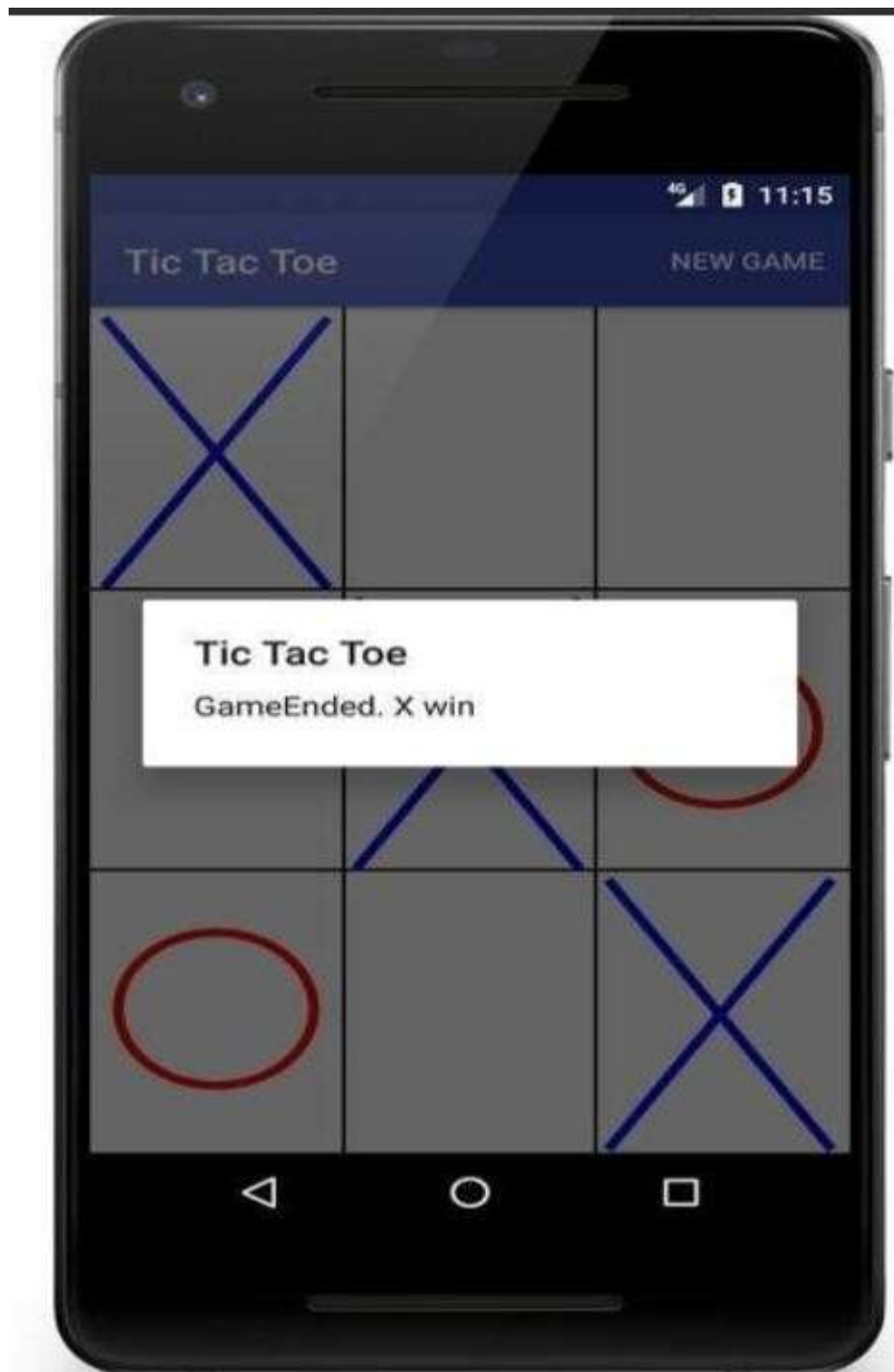
```
setOnDismissListener(new DialogInterface.OnDismissListener() {  
    @Override public void onDismiss(DialogInterface  
    dialogInterface) { newGame();  
    }  
}).show();  
}  
  
private void newGame() {  
    gameEngine.newGame(); boardView.invalidate();  
    }  
}
```

Output:



Playing to the Tic-Tac-Toe Game

The game works great and finally, we win the game against the computer which is logical because our Artificial Intelligence (AI) is really basic:



Result : Thus the Tic-tac-Toe game application was developed successfully.

Experiment no:08

Name:Dayanand Jagdale Roll no:42

```
<definitions xmlns="http://ws.apache.org/ns/synapse">
  <proxy name="Stock"Proxy"
    transports="http"
    startOnLoad="true" trace="disable">
      <target>
        <endpoint> <address
          uri="jms:/SimpleStock"Service?transport.jms.ConnectionFactoryJNDIName=TopicConnection
          Factory<ampampjava.naming.factory.initial=org.apache.activemq.jndi.ActiveMQInitialContextFa
          ctory
          <ampampjava.naming.provider.url=tcp://localhost:64242<ampamptransport.jms.DestinationTy
          pe=topic"/< </endpoint>
        <inSequence>
          <property name="OUT_ONLY" value="true"/>
        </inSequence>
        <outSequence>
          <send/>
        </outSequence>
      </target>
    </proxy>
  </definitions>
```

Configuring the subscribers

```
<definitions xmlns="http://ws.apache.org/ns/synapse">
```

```
<proxy name="SimpleStock"eService1"
  transports="jms" startOnLoad="true"
  trace="disable">
  <description>
  <target>
  <inSequence>
  <property name="OUT_ONLY" value="true">
  <log level="custom">
  <property name="Subscriber1" value="I am Subscriber1">
  <log>
  <drop>
  <inSequence>
  <outSequence>
  <send>
  <outSequence>
  <target>
  <parameter name="transport.jms.ContentType">
  <rules>
  <jmsPropertycontentType/jmsProperty>
  <defauapplication/xml/defau>
  <rules>
  <parameter>
  <parameter name="transport.jms.ConnectionFactory"myTopicConnectionFactory/parameter>
  <parameter name="transport.jms.DestinationType"topic/parameter>
  <parameter name="transport.jms.Destination"SimpleStock"eService/parameter>
  <proxy>
```

```

<proxy name="SimpleStock"eService2">
  transports="jms" startOnLoad="true"
  trace="disable">
    <description
    <target>
    <inSequence>
    <property name="OUT_ONLY" value="true"/>
    <log level="custom"/>
    <property name="Subscriber2" value="I am Subscriber2"/>
    <log>
    <drop>
    <inSequence>
    <outSequence>
    <send>
    <outSequence>
    <target>
    <parameter name="transport.jms.ContentType"/>
    <rules>
    <jmsPropertycontentTypejmsProperty>
    <defauapplicationxmldefau>
    <rules>
    <parameter>
    <parameter name=<transport.jms.ConnectionFactory>myTopic<ConnectionFactoryparameter>
    <parameter name=<transport.jms.DestinationType>topic<parameter>
    <parameter name=<transport.jms.Destination>SimpleStock<eServiceparameter/>
    <proxy>

```

<definitions>

Next, you configure two proxy services that subscribe to the JMS topic SimpleStockQuoteService, so that whenever this topic receives a message, it is sent to these subscribing proxy services. Following is the sample configuration for these proxy services.

Publishing to the topic

Start the ESB with one of the following commands:

<ESB_HOME>/bin/wso2server.sh (on Linux)

<MB_HOME>/bin/wso2server.bat (on Windows)

A log message similar to the following will appear:

INFO {org.wso2.andes.server.store.CassandraMessageStore} - Created Topic :

SimpleStockQuoteService

INFO {org.wso2.andes.server.store.CassandraMessageStore} -

Registered Subscription tmp_127_0_0_1_44759_1 for Topic SimpleStockQuoteService

To invoke the publisher, use the sample stockquote client service by navigating to

<ESB_HOME>/samples/axis2Client and running the following command:

```
ant stockquote -Daddurl=http://localhost:8280/services/StockQuoteProxy -Dmode=placeorder -
Dsymbol=MSFT
```

The message flow is executed as follows:

When the stockquote client sends the message to the StockQuoteProxy service, the publisher is invoked and sends the message to the JMS topic.

The topic delivers the message to all the subscribers of that topic. In this case, the subscribers are

ESB proxy services.

Result : Implementing Publish/Subscribe Paradigm using Web Services, ESB and JMS is successfully executed

Experiment no:09

Name:Dayanand Jagdale Roll No:42

Minimal WSDL (interface + resource properties)

```
<?xml version="1.0" encoding="UTF-8"?> <definitions
name="StickyNoteService"
targetNamespace="http://example.org/stickynote"
xmlns:tns="http://example.org/stickynote"
xmlns:wsrp="http://docs.oasis-open.org/wsrp/rp-2"
xmlns:soap="http://schemas.xmlsoap.org/wsdl/soap/"
xmlns="http://schemas.xmlsoap.org/wsdl/">

  <!-- Types: resource property schema -->

  <types>

    <xsd:schema xmlns:xsd="http://www.w3.org/2001/XMLSchema"
targetNamespace="http://example.org/stickynote/types">

      <xsd:element name="message" type="xsd:string"/>

    </xsd:schema>

  </types>

  <!-- PortType -->

  <portType name="StickyNotePortType">

    <operation name="setMessage">

      <input message="tns:setMessageRequest"/>

      <output message="tns:setMessageResponse"/>

    </operation>

  </portType>

</definitions>
```



```

</operation>
<operation name="getMessage">
  <input message="tns:getMessageRequest"/>
  <output message="tns:getMessageResponse"/>
</operation>
</portType>
<!-- Messages (brief) -->
<message name="setMessageRequest">
  <part name="message" element="tns:message"/>
</message>
<message name="setMessageResponse">
  <part name="status" type="xsd:string"/>
</message>
<message name="getMessageResponse">
  <part name="message" element="tns:message"/>
</message>
<!-- Abstract: WSRF ResourceProperties declaration (conceptual) -->
<!-- In GT4 examples there are extra resource descriptor / schema files; adapt per tutorial. -->
</definitions>

```

Java implementation (skeleton)

```

package org.example.stickynote; import
org.globus.wsrp.Resource; import
org.globus.wsrp.ResourceHome; import

```

```

org.globus.wsrp.ResourceContext;
import javax.xml.namespace.QName;

public class StickyNoteServiceImpl {

// This is the method exposed by WSDL

public String setMessage(String
message) throws Exception {

    // retrieve resource for this invocation

    StickyNoteResource res = (StickyNoteResource)
ResourceContext.getResourceContext().getResource();
res.setMessage(message);    return "OK";

}

public String getMessage() throws Exception {

    StickyNoteResource res = (StickyNoteResource)
ResourceContext.getResourceContext().getResource();
return res.getMessage();

}

}

```

And a simple resource class (holds the state):

```

package org.example.stickynote;

public class StickyNoteResource /* implements Resource or extends BaseResource as per GT4
samples */ {    private String message = "";    public synchronized void setMessage(String m)
{        this.message = m;

        // if you need persistence, write out here

    }

    public synchronized String getMessage() {
return message;

```

```
}  
}
```

WSDD (Axis deployment descriptor) — example services.wsdd

```
<deployment xmlns="http://xml.apache.org/axis/wsdd/"  
  xmlns:java="http://xml.apache.org/axis/wsdd/providers/java">  
  <service name="StickyNoteService"  
    provider="java:RPC">  
    <parameter name="className" value="org.example.stickynote.StickyNoteServiceImpl"/>  
    <parameter name="allowedMethods" value="setMessage,getMessage"/>  
    <requestFlow>  
      <!-- GT4 adds handlers (security, wsrf) when deployed in container -->  
    </requestFlow>  
  </service>  
</deployment>
```

Ant build.xml to produce .gar

```
<project name="StickyNote" default="dist" basedir=".">  
  <property name="src.dir" value="src"/>
```

```
<property name="build.dir" value="build"/>
<property name="lib.dir" value="lib"/>
<property name="dist.dir" value="dist"/>
<property name="gar.name" value="StickyNote.gar"/>
<target name="clean">
    <delete dir="${build.dir}"/>
    <delete dir="${dist.dir}"/>
</target>
<target name="compile">
    <mkdir dir="${build.dir}"/>
    <javac srcdir="${src.dir}" destdir="${build.dir}" includeantruntime="false">
        <classpath>
            <fileset dir="${lib.dir}">
                <include name="**/*.jar"/>
            </fileset>
        </classpath>
    </javac>
</target>
<!-- copy resources: WSDL, WSDD, resource descriptor, META-INF -->
<target name="copy-resources" depends="compile">
    <mkdir dir="${build.dir}/META-INF"/>
    <copy file="services.wsdd" todir="${build.dir}/META-INF"/>
    <copy file="StickyNote.wsdl" todir="${build.dir}"/>
    <copy file="resource-descriptor.xml" todir="${build.dir}"/>
    <!-- include any xsds -->
    <copy todir="${build.dir}/schema">
```

```
<fileset dir="schema" includes="**/*.xsd"/>
</copy>
</target>
<target name="gar" depends="copy-resources">
  <mkdir dir="${dist.dir}"/>
  <jar destfile="${dist.dir}/${gar.name}" basedir="${build.dir}">
    <manifest>
      <attribute name="Created-By" value="Ant"/>
    </manifest>
  </jar>
</target>
<target name="dist" depends="gar"/>
</project>
```

Output:

Ant build output (successful ant -f build.xml)

Buildfile: /home/user/stickynote/build.xml

clean:

[delete] Deleting directory /home/user/stickynote/build

[delete] Deleting directory /home/user/stickynote/dist

compile:

[mkdir] Created dir: /home/user/stickynote/build

[javac] Compiling 4 source files to /home/user/stickynote/build

[copy] Copying 1 file to /home/user/stickynote/build

[copy] Copying 1 file to /home/user/stickynote/build/schema

copy-resources:

[copy] Copying 1 file to /home/user/stickynote/build/META-INF

[copy] Copying 1 file to /home/user/stickynote/build

[copy] Copying 1 file to /home/user/stickynote/build

[copy] Copying 2 files to /home/user/stickynote/build/schema

gar:

[mkdir] Created dir: /home/user/stickynote/dist

[jar] Building jar: /home/user/stickynote/dist/StickyNote.gar

BUILD SUCCESSFUL

Total time: 3 seconds

.gar contents (what the GAR archive contains)

META-INF/

META-INF/services.wsdd StickyNote.wsdl

schema/message.xsd resource-descriptor.xml org/

org/example/ org/example/stickynote/

org/example/stickynote/StickyNoteServiceImpl.class

org/example/stickynote/StickyNoteResource.class

org/example/stickynote/ResourceHomeImpl.class

META-INF/MANIFEST.MF

GT4 container deployment log (snippet)

Oct 10 14:02:11 INFO DeploymentService - Received deploy request:
/path/to/dist/StickyNote.gar

Oct 10 14:02:11 INFO GARDeployer - Extracting GAR: StickyNote.gar

Oct 10 14:02:12 INFO ServiceDeployer - Deploying service: StickyNoteService

Oct 10 14:02:12 INFO AxisServiceFactory - Deploying Axis service 'StickyNoteService' Oct

10 14:02:12 INFO WSRFProvider - Installing WSRF handlers for StickyNoteService

Oct 10 14:02:12 INFO ResourceHomeManager - Resource Home registered:
StickyNoteService/home

Oct 10 14:02:12 INFO AdminService - Service StickyNoteService deployed successfully.

Oct 10 14:02:12 INFO Endpoint - Service available at:

<https://grid.example.org:8443/services/StickyNoteService?wsdl> **Example resource creation /
EPR (when a client creates or the service auto-creates a resource)**

EPR:

<EndpointReference>

<Address><https://grid.example.org:8443/services/StickyNoteService/resource/123e4567e89b-12d3-a456-426614174000></Address>

<ReferenceProperties>

<ResourceKey>

<ServiceName>StickyNoteService</ServiceName>

<ResourceID>123e4567-e89b-12d3-a456-426614174000</ResourceID>

```
</ResourceKey>
</ReferenceProperties>
</EndpointReference>
```

Example SOAP request & response — setMessage

POST /services/StickyNoteService/resource/123e4567-e89b-12d3-a456-426614174000 HTTP/1.1

Host: grid.example.org:8443 Content-Type:

text/xml; charset=utf-8

Content-Length: ...

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/"
  xmlns:tns="http://example.org/stickynote">
  <soapenv:Header/>
  <soapenv:Body>
    <tns:setMessage>
      <tns:message>Hello, this is my sticky note!</tns:message>
    </tns:setMessage>
  </soapenv:Body>
</soapenv:Envelope>
```

SOAP Response (setMessage):

HTTP/1.1 200 OK

Content-Type: text/xml; charset=utf-8


```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/">
  <soapenv:Body>
    <ns:setMessageResponse xmlns:ns="http://example.org/stickynote">
      <status>OK</status>
    </ns:setMessageResponse>
  </soapenv:Body>
</soapenv:Envelope>
```

Example SOAP request & response — getMessage

POST /services/StickyNoteService/resource/123e4567-e89b-12d3-a456-426614174000 HTTP/1.1

Host: grid.example.org:8443

Content-Type: text/xml; charset=utf-8

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/"
  xmlns:tns="http://example.org/stickynote">
  <soapenv:Header/>
  <soapenv:Body>
    <tns:getMessage/>
  </soapenv:Body>
</soapenv:Envelope>
```

SOAP Response (getMessage):

HTTP/1.1 200 OK

Content-Type: text/xml; charset=utf-8

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/">
  <soapenv:Body>
    <ns:getMessageResponse xmlns:ns="http://example.org/stickynote">
      <message>Hello, this is my sticky note!</message>
    </ns:getMessageResponse>
  </soapenv:Body>
</soapenv:Envelope>
```

Sample client console output (Java client invoking the service)

Creating StickyNote resource...

Resource created. EPR =

<https://grid.example.org:8443/services/StickyNoteService/resource/123e4567-e89b-12d3a456-426614174000>

Calling setMessage("Hello, this is my sticky note!")...

setMessage returned: OK

Calling getMessage()...

getMessage returned: Hello, this is my sticky note!

Client finished.

If persistence is enabled (for example resource state persisted to disk/db)

Oct 10 15:10:02 INFO ResourcePersistenceManager - Restored resource ResourceID=123e4567-... from /var/globus/resources/123e4567-....ser

Oct 10 15:10:02 INFO StickyNoteResource - message property restored: "Hello, this is my sticky note!"